Steering SAE Features for Content-Preserving Style Control in Diffusion Models

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Abstract

User-defined generative prompts regularly conflate content and style attributions in image diffusion frameworks like Stable Diffusion 1.4 (SD1.4). Often, style perturbs content meant to be preserved by model layers. The resulting embedding space entangles features. Style-transfer failures are historically mechanistic in diffusion. This paper investigates how steering interpretable sparse autoencoder (SAE) features inside the model U-Net can apply style at adjustable strengths, in isolation, while leaving pure content-specific prompts intact. We train a late-block SAE for graded injection on SD1.4 and localize style features by cross-style contrast over matched-anchor prompts; their summed top-τ decoder directions, added to the CFG conditional branch, are amplified by guidance. On held out prompts, we score style through the UnlearnCanvas ViT classifier and content through CLIP cosine similarities across images and the style-free baseline. Steering preserves far more content than prompting (clip content 0.876 vs 0.772) while tracing a continuous style–content frontier prompting cannot. Style fidelity is metric-dependent: prompting leads on clip style (0.249 vs 0.218), though the two style metrics are uncorrelated ($r = 0.007$). The dominant finding is per-style heterogeneity, as steering succeeds for near-monosemantic styles (Watercolor) but fails for line-dominated ones (Pencil Drawing).

1. Introduction

Sparse autoencoders (SAEs) are widely used interpretation tools for large neural net activations. They decompose selections into large latent spaces of approximately monosemantic features. Amplifying activations via dictionary elements in the SAE can steer model behavior [1, 12, 6]. In text-to-image diffusion, SAEs trained on U-Net activations recover concept-aligned visual features [11]. Previous work has demonstrated that an exploitation of SAEs can erase a target concept from the model [2]. Erasure, however, is binary. Image generation requires rendering attributes and activations at chosen intensities. State of the art models allow for user control through prompts, singular conditioning strings, where style and content attributions are interwoven through tokens. Content and style therefore interact through cross-attention at every U-Net layer. Strengthening style attributions in the prompt dilutes the scene it should likely preserve. [7, 13]. Following this result, we inquire into whether SAE features that enable concept removal can instead enable graded control. We hypothesized that steering activations via identified SAE style features produces similar style classification, while, at the same time, disentangles style from content quantitatively more than prompting.

To investigate this, our project aims to:

- Localize the SAE features encoding a given style without labels, by contrasting matched-anchor prompts across styles.
- Steer those features at inference to inject style at controllable strength
- Identify the regime under which injection succeeds
- Quantify style-versues-content trade-offs; benchmark sparse-feature steering against prompt conditioning

Motivations are both practical and scientific. In practice, the prompt affords no smooth control. Style tokens act as discrete switches entangled with the random seed. An attribution cannot be turned up or down without re-generating the composition, and existing concept-removal methods operate all-or-nothing [4, 9]. Representation-level handling that touches only style-bearing features minimizes this, in theory, with direct applcations to creative editing, generation, and content-preserving suppression of styles. Scientifically, the same question probes how diffusion frameworks encode style. Whether small sets of sparse features carry style encodings cleanly tests whether linear, near-monosemantic representations observed in language models extend to generative vision. Outcomes of this paper will bound how precisely style can be steered and how interpretable, separable, these models' internal feature representations are. Across all styles, steering preserves substantially more content than prompting (`clip_content`

0.876 vs. 0.772) and traces a continuous style content frontier prompting cannot, while style fidelity is metric-dependent: prompting leads on `clip_style` and steering is nominally higher on `uc_style`, with the uniform `injected_normal` variant strongest overall.

2. Related Work

Our investigation into disentanglement via SAE-feature steering draws upon several areas of deep learning and generative vision research:

1. **Unlearning in Diffusion.** Historically, concept ablation has fine-tuned models to map a target concept onto an anchor distribution. This curates per-class anchor prompts [9]; Additionally, erased Stable Diffusion edits attention weights with negative guidance [4]. Both end up modifying weights globally and reveal little about the internal changes. UnlearnCanvas standardizes evaluation with a high-resolution stylized dataset and automatic style classifiers across this research domain [14]. SAeUron brings out novelty as it avoids fine-tuning entirely. SAEs, trained on U-Net activations across denoising steps, can provide sparse features in the latent space tied to visual concepts. The work goes on to ablate them at inference [2]. We adopt some of SAeUron's machinery and feature-localization, but after training an SAE on SD1.4 late-block layers, we convert ablation into additive steering.

2. **Sparse autoencoders and superposition.** Networks place more features than neurons in superposition [3]; sparse dictionary learning recovers an over-complete, rather interpretable basis [1]. We use the top-k formulation, which fixes sparsity directly instead of tuning an L_1 penalty [10, 6], with the BatchTopK variant that selects the top activations across a batch during training. It continues to keep per-sample top-k at inference. The SAE will use an auxiliary loss to reconstruct the residual from dead latents to keep the dictionary, latent space, alive [6].

3. **Activation steering.** A parallel line edits behavior by intervening on representations rather than weights or inputs. In diffusion, concept directions can be added to activations and exposed as continuous "sliders" [5]. Our steering is an instance of this performed in the SAE basis, sparse, where directions correspond to individual decoder atoms. This avoids dealing with dense, polysemantic vectors.

4. **Style and denoising.** Cross-attention maps largely fix the prompt-to-region correspondence. Thus, the entanglement we target also remains fairly constant [7]. "Resolution chromatography" characterizes denoising as coarse-to-fine, with high-frequency details emerging late [8]. Since painterly style is largely high-frequency texture, this motivates intervening on the decoder, up-sampling path. This is consistent with SAeUron's finding that the block `up.1.2` specializes in stlye [2].

3. Data

Our data comes in three form factors: the text-prompt corpus, the U-Net activations cached when running Stable Diffusion 1.4 on prompts, and the images rendered for evaluation. The activations, training data used for the SAE, are thus produced by the model itself.

We test and build our prompt corpus on the anchor prompts of Kumari *et al*. [9]. This includes 80 authored GPT-generated content variations per object class, each one-sentence descriptions of scenes, activities, and settings. Prior work has historically trained Concept Ablation methods on this data; across 20 object classes this gives $1,600$ prompts. Furthermore, these prompts were adopted as the neutral set by the UnlearnCanvas benchmark [14] and reused with style suffixes, for SAE training in SAeUron [2]. We follow the same lineage for comparability. Styles in the entangled prompt set are drawn from UnlearnCanvas's catalogue of 50 artistic styles, of which we use 10 visually distinct ones (Van Gogh, Monet, Picasso, Cubism, Watercolor, pop Art, Cartoon, Ukiyo-e, Pencil Drawing, Impressionism). A styled prompt is formed as $\langle$*anchor*$\rangle$ *in* $\langle$*style*$\rangle$ *style*.

For SAE training, we sample 400 anchors per style across the ten. Accordingly, we use $4,000$ styled prompts. Running SD1.4's 30-step denoising loop on each (with the VAE decode skipped for emphasis on U-Net forwards), we hook the late upsampling block `up.1.2`, whose output is a $[1, 1280, 16, 16]$ tensor. Following SAeUron and previous literature, we keep only the text-conditioned CFG half and flatten each denoising step into 256 spatial rows of dimension 1280. This yields $\sim$7.7k rows per prompt and $\sim$30M rows in total. This is streamed to disk in fp16 from a single AWS g6.xlarge.instance.

Two disjoint prompt sets isolate localization from evaluation.

1. *Feature-localization.* This set contains 20 entangled prompts per style (one per object class) in which every style reuses the same anchor. Contrasting activations across styles therefore isolates style-specific features from content.

2. *Inference.* This set contains 20 neutral content prompts per style (no style word) drawn from anchor rows disjoint from the localization set, so features are never evaluated on the prompts used to select them.

All evaluation images are rendered at 512×512 with 30 denoising steps and classifier-free-guidance scale 7.5. For each content prompt we also render a style-free baseline (identical prompt and seed with no style word) that serves as the preservation reference.

4. Methods

To overview our pipeline's four main stages at a high level, we: 1) collect and cache the SD1.4 `up.1.2` activations described in Section 3 and train a Top-k autoencoder on them, 2) identify the latents that encode a target style by contrasting styled activations against neutral activations, 3) re-inject those latents as additive decoder directions on the conditional branch at inference with controllable strength, and 4) score the rendered images for style presence and content preservation.

4.1. Sparse Autoencoder

Let $x \in \mathbb{R}^d$, d=1280, describe a single flattened spatial row of the `up.1.2` feature map on the text-conditioned portion of the CFG batch. We train a top-k SAE [10, 6] that projects x into an over-complete high-dimensional latent space and subsequently reconstructs it from a unit-normalized dictionary,

$$z = \mathrm{TopK}\big(W_{\mathrm{enc}}(x - b_{\mathrm{dec}})\big), \qquad \hat{x} = W_{\mathrm{dec}}\, z + b_{\mathrm{dec}}, \quad (1)$$

where $\mathrm{TopK}(\cdot)$ keeps the k largest pre-activations and zeroes the rest, thereby fixing the L_0 sparsity directly and addressing the shrinkage bias of the alternative L_1 penalty approach [6]. We use an expansion factor of 16, which gives n=20480 latents, sparsity k=32, decoder rows constrained to unit norm, and a learned pre-encoder bias b_{dec} that centers the input. These choices mirror the SAE configuration developed for the same U-Net block in SAeUron [2].

During training, we implement BatchTopK, a relaxation on conventional TopK. Rather than enforcing k active latents per row, we keep the largest $k \times B$ pre-activations across the whole batch of B rows, allowing a variable number of active latents per sample. At inference, we revert to a fixed per-row top-k=32. We deliberately use per-row TopK at inference rather than the canonical BatchTopK choice because this keeps the inference-time sparsity identical to the steering budget. This diverges from previous work.

In order to keep the dictionary alive, we add an AuxK term [6], such that a latent is flagged dead if it has not fired in the last 10M tokens. The auxiliary loss reconstructs the residual $e = x - \hat{x}$ from the top-k_{aux} dead latents, with $k_{\mathrm{aux}} \approx d/2$. The training objective becomes:

$$\mathcal{L} = \|x - \hat{x}\|_2^2 \;+\; \alpha\, \|e - \hat{e}_{\mathrm{aux}}\|_2^2, \qquad \alpha = \tfrac{1}{32}, \quad (2)$$

where $\hat{e}_{\mathrm{aux}}$ is the dead-latent reconstruction of e. The same α=1/32 is used for diffusion SAEs in Concept Steerers [**?**].

We optimize with Adam at learning rate $4\mathrm{e}{-4}$ and batch size 4096 for 10 epochs over the $\sim$30M cached rows. Reconstruction quality is reported as the fraction of variance unexplained,

$$\mathrm{FVU} = \frac{\mathbb{E}\,\|x - \hat{x}\|_2^2}{\mathbb{E}\,\|x - \bar{x}\|_2^2}, \quad (3)$$

i.e. the normalized MSE; we reach FVU$\approx$0.17, comparable to the 0.198 reported by SAeUron at `up.1.2` [2].

4.2. Style-Feature Identification

We isolate the latents carrying a style s without any latent-level labels by identifying activations' contrasts over matched-anchor prompts (Section 3, feature-localization set). For every latent i, we compute its mean activation over the styled rows of s and over the neutral rows, and take the difference

$$\Delta_i^s = \bar{f}_i^{\,s} - \bar{f}_i^{\,\varnothing}. \quad (4)$$

The intuition here is simply that the most divergent activations in the over-complete SAE latent space will correspond to the style's interpretable linear directions. Because the styled and neutral prompts share the same anchors and differ only by the appended style suffix, content-bearing latents cancel in Δ^s and style-bearing latents survive. We keep the latents above the 99.9th percentile of Δ^s as the style set $\mathcal{F}_s$. For the patch-level variant, we additionally restrict at each spatial row p to the top-16 active latents at that row, intersected with $\mathcal{F}_s$.

4.3. Inference-Time Steering

We turn the localized latents into a steering direction by summing their (unit-norm) decoder atoms,

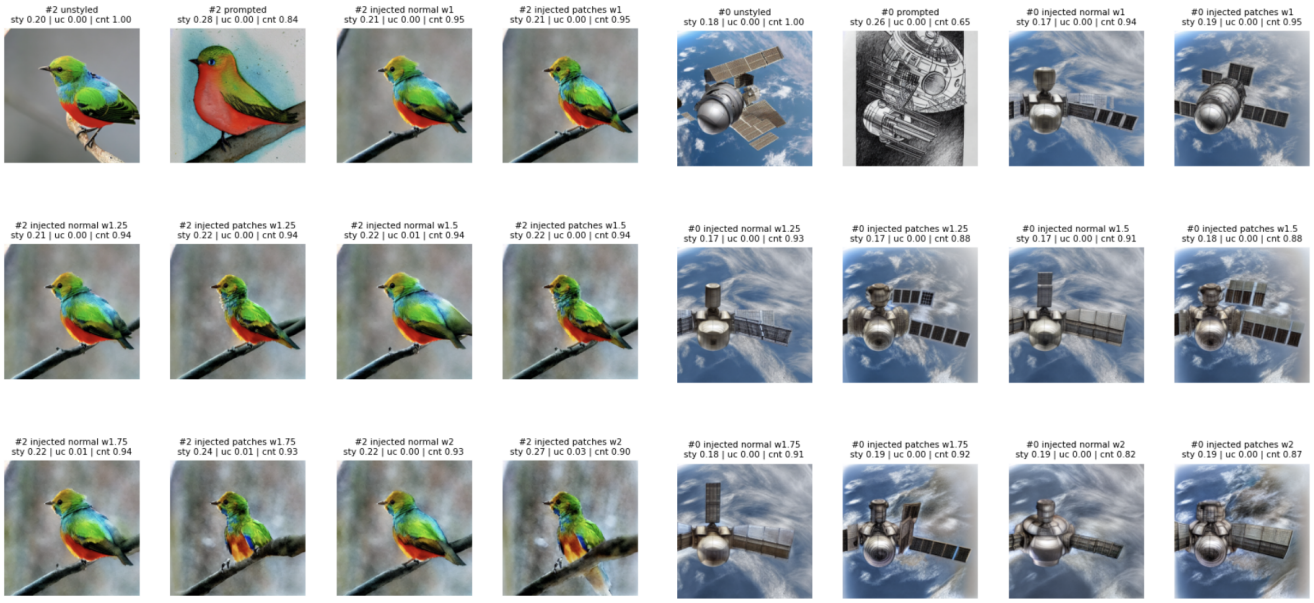
$$v_s = \sum_{i \in \mathcal{F}_s} W_{\mathrm{dec}}^{(i)}, \quad (5)$$

and add it to the conditional activation at every denoising step. We employ two steering procedures, termed the *uniform* and *patch* variants. In the *uniform* variant, the same v_s is added to all 256 spatial rows, while in the *patch* variant, the sum is taken per row over the style latents active at that row:

$$\text{uniform:} \quad x_{t,p}^c \leftarrow x_{t,p}^c + w\, v_s, \quad (6)$$

$$\text{patch:} \quad x_{t,p}^c \leftarrow x_{t,p}^c + w \sum_{i \in \mathcal{F}_s \cap \mathrm{Top16}(p)} W_{\mathrm{dec}}^{(i)}, \quad (7)$$

where w is the user-controllable steering strength and $x_{t,p}^c$ is the row-p activation on the conditional branch at step t. This is the textbook additive activation-steering operator $R' = R + v$ [**?**], realized in the SAE basis so that the direction is a sum of near-monosemantic decoder atoms rather than a dense, polysemantic vector. Concept Steerers apply the same construction in a diffusion SAE [**?**].



a Watercolor: a steerable (near-monosemantic) style.

b Pencil Drawing: a style steering cannot render.

Figure 1: **Per-style heterogeneity.** A single prompt/seed shown unstyled, prompted, and under both steering variants across w ranging from 1 to 2. (a) style intensifies smoothly while content holds; (b) steering inadequately injects the desired style. metric legend: `sty=clip_style`, `uc=uc_style`, `cnt=clip_content`.

Applying steering only on the conditioned branch of CFG lets guidance do the amplification for free. With guidance scale g=7.5, the sampler forms:

$$\hat{\epsilon}_t = \epsilon_\theta(x_t, \varnothing) + g\big(\tilde{\epsilon}_\theta(x_t, c) - \epsilon_\theta(x_t, \varnothing)\big), \qquad (8)$$

where $\tilde{\epsilon}_\theta(x_t, c)$ is the noise prediction computed from the steered conditional activations [**?**]. Because the unconditional branch is untouched, the injected direction enters the extrapolation $g(\tilde{\epsilon} - \epsilon^{\varnothing})$ and is effectively scaled by g; the realized strength is thus $\sim g\,w$. We sweep $w \in \{1, 2\}$ to trace the style–content frontier.

4.4. Evaluation

We score each inference image (Section 3) on three axes. Style presence is measured with the reference-free CLIP-Score similarity [**?**] between the image and the style descriptor, using CLIP ViT-B/32,

$$\texttt{clip_style} = \cos\big(\phi_{\text{img}}(I),\ \phi_{\text{txt}}(\text{“}\ldots\text{in } s \text{ style”})\big). \qquad (9)$$

Content preservation is measured against the per-prompt style-free baseline I_{base} (identical prompt and seed, no style word) as an image-to-image CLIP cosine,

$$\texttt{clip_content} = \cos\big(\phi_{\text{img}}(I),\ \phi_{\text{img}}(I_{\text{base}})\big), \qquad (10)$$

which is 1.0 by construction for the baseline itself. Finally, `uc_style` is the score the UnlearnCanvas ViT-L 51-way style classifier assigns to the target style [14]. We caution that this classifier is calibrated to UnlearnCanvas's fine-tuned-SD appearance rather than steered base SD1.4, so absolute `uc_style` values are low across all conditions and are most informative as relative comparisons. We report means over the 10 styles $\times$ 20 inference prompts and benchmark sparse-feature steering, at both strengths and both spatial variants, against ordinary prompt conditioning (the ⟨*anchor*⟩ *in* ⟨*style*⟩ *style* baseline).

5. Results

5.1. Setup

We evaluate SAE steering against text prompting on the 10 styles $\times$ 20 neutral inference prompts of Section 3 (200 image pairs per condition, fixed seed 42). Each image is scored on two style-fidelity metrics (`clip_style` and `uc_style`), and two content-fidelity metrics (`clip_content` and `clip_prompt`), described at greater length in Section 4.4. We compare three regimes: unstyled (⟨prompt⟩), prompted(⟨prompt⟩ in ⟨style⟩ style), and injected (SAE steering), the last in the uniform (`injected_normal`) and patch (`injected_patches`) variants of Section 4, swept over steering weights $w \in [1, 2]$.

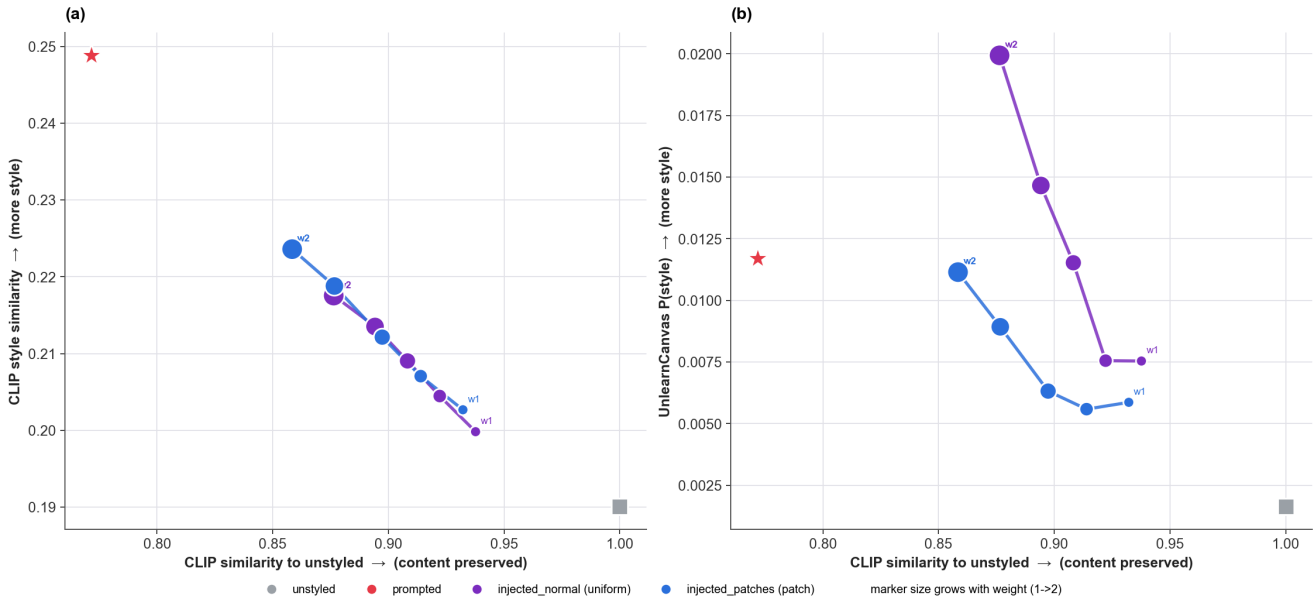


Figure 2: **Style↔content trade-off.** Left: `clip_style` vs. content preservation; right: `uc_style` vs. content. Marker size grows with steering weight (w=1→2); up-and-left is better. In CLIP space the prompted point (star) lies essentially on the linear extrapolation of the uniform steering frontier, whereas on the classifier uniform-$w2$ mean-dominates it (further up *and* right).

5.2. Qualitative analysis

We begin with an insightful qualitative analysis of steering performance on ntwo different styles (watercolor and pencil drawing). Figures 1a and 1b each illustrate a single prompt subjected to unstyled, prompted, and steered conditions across a range of weights, and they expose a sharp split between styles our intervention can render and styles it cannot. For Watercolor (Fig. 1a) steering behaves as intended: as w rises, soft edges, pigment pooling, and paper-grain texture intensify monotonically while the subject's pose, color layout, and composition stay nearly fixed. At the strongest patch setting, the style score reaches 0.27, approaching prompting's 0.28. Yet, content holds at 0.90 against prompting's 0.84. This is indicative of the expected SAE activation behavior that informed our approach: style manifests in approximately monosemantic, steerable activations. However, the Pencil Drawing (Fig. 1b) exhibits the opposite. Prompting yields a convincing graphite sketch (`clip_style`=0.26), but steering never reaches that compelling appearance. Quantitatively, across the full range, the style score barely moves (0.17–0.19, at or below the unstyled 0.18). Worse, increasing the weight adds no hatching and does not desaturate the image. The identified SAE directions perturb shape rather than impose a line drawing, as expected if pencil drawing's defining cues (high-frequency line structure, absence of color) are *not* captured by a compact feature set at the block we steer.

5.3. Content preservation

It is evident that our steering approach retains far more of both the underlying image and the prompt than prompting does by CLIP cosine similarity. Pairwise (steered − prompted), `clip_content` improves by $+0.105\pm0.014$, with steering ahead in 81.5% of the 200 pairs, and `clip_prompt` improves by $+0.013\pm0.004$ (steering ahead in 68%). The effect is universal, as steering preserves more content than prompting in all 10 styles with the per-style gap ranging from $\approx+0.05$ for Impressionism to $\approx+0.18$ for Cubism (Fig. 5). In short, the style suffix imposes a large content score reduction, pulling `clip_content` from 1.0 to 0.772, whereas steering reaches a styling regime while holding content near 0.876 (Table 1; Fig. 3).

5.4. Style fidelity

With respect to style, however, the picture is mixed, and the two style metrics are essentially uncorrelated across images (Pearson r=0.007; Fig. 4). Therefore, the conclusion depends on the measure. On `clip_style`, prompting performs better than steering (0.249 vs 0.218; paired $\Delta=-0.031\pm0.005$, steering ahead in only 18% of pairs and lower in all 10/10 styles; Fig. 3). This gap is partly a confound: `clip_style` rewards literal CLIP alignment to the same style tokens from which the prompted images were generated. On `uc_style`, steering is nominally higher

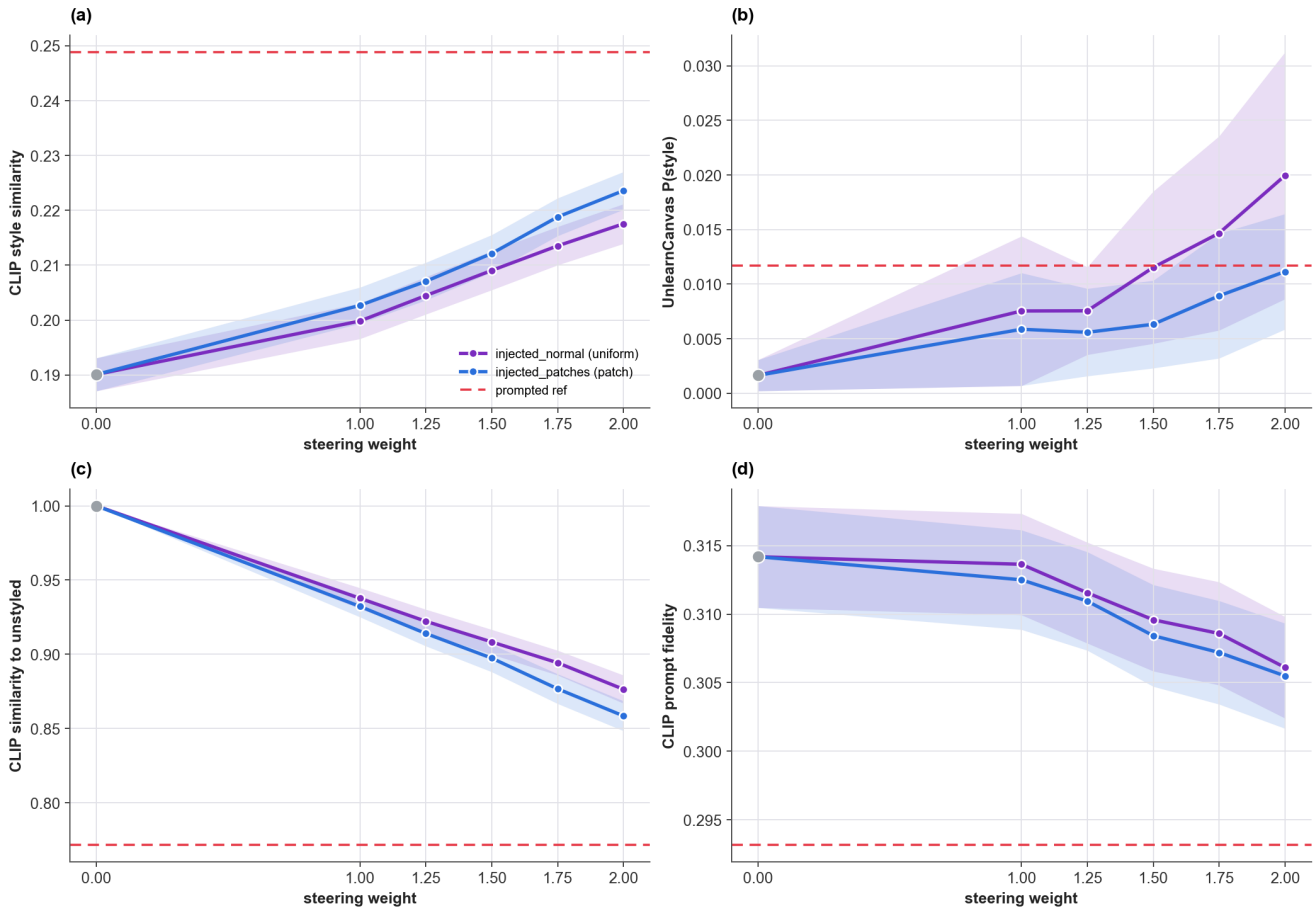


Figure 3: **Each metric vs. steering weight** (mean $\pm 95\%$ CI over 10 styles $\times$ 20 prompts). Style metrics rise and content metrics fall monotonically with w; the prompted reference is dashed. Steered `clip_style` stays below the prompted line at every weight, while steered content stays far above it.

Condition	`clip_style`↑	`uc_style`↑	`clip_content`↑	`clip_prompt`↑
unstyled	0.190 ± 0.003	0.0016	1.000	0.314 ± 0.004
prompted	$\mathbf{0.249} \pm 0.004$	0.0117 ± 0.0053	0.772 ± 0.013	0.293 ± 0.004
injected (w=2)	0.218 ± 0.004	$\mathbf{0.0199} \pm 0.0113$	$\mathbf{0.876} \pm 0.010$	$\mathbf{0.306} \pm 0.004$

Table 1: **Mean scores** over the 200 prompt×style pairs (uniform steering at w=2 as the representative operating point). Higher `clip_style`/`uc_style` indicates more style; higher `clip_content`/`clip_prompt` indicates more preserved content. Intervals are 95% confidence intervals and bold marks the best non-trivial style score per metric.

(0.0199 vs 0.0117), but the paired difference is not significant (Δ=+0.0082±0.0126, win rate 51.5%) and is positive in only 4/10 styles (clearest for Monet, Watercolor, Pop Art, and Cubism). Steering only scores higher on both `uc_style` and `clip_content` in 37% of pairs.

5.5. Style–Content Tradeoff

The steering weight traces a smooth style–content frontier (Fig. 2, 3). As w rises form 1.0 to 2.0, `clip_content` falls monotonically (from 0.94 to 0.88 for uniform) while both style metrics rise monotonically. This continuous control is the qualitative payoff that prompting, which essentially discretely activates/inactivates style with these seeds, cannot provide.

The stronger claim that steering buys style more cheaply than prompting requires discussion as well. On `clip_style` it does not: the five uniform operating points lie on a near-linear style–content frontier, and a linear fit through them, extrapolated to prompting's content level of 0.772, predicts `clip_style`≈0.249, which is indistin-

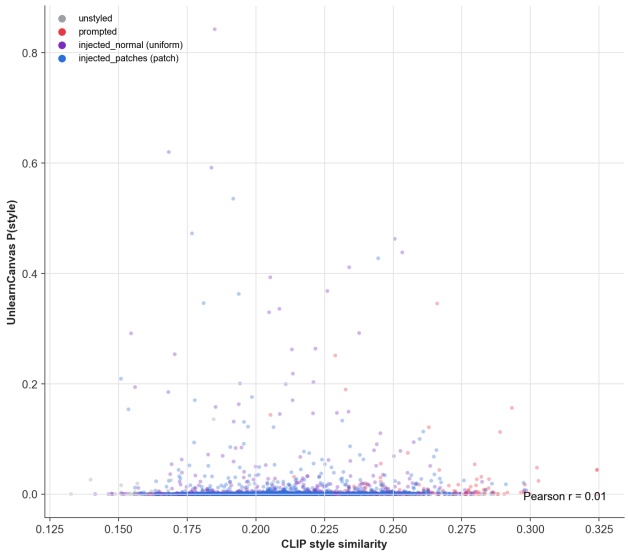


Figure 4: **Do the two style metrics agree?** `clip_style` vs. `uc_style` for every image. The correlation is negligible (Pearson r=0.007), so the two measures capture different notions of "style" and we report both rather than privileging one.

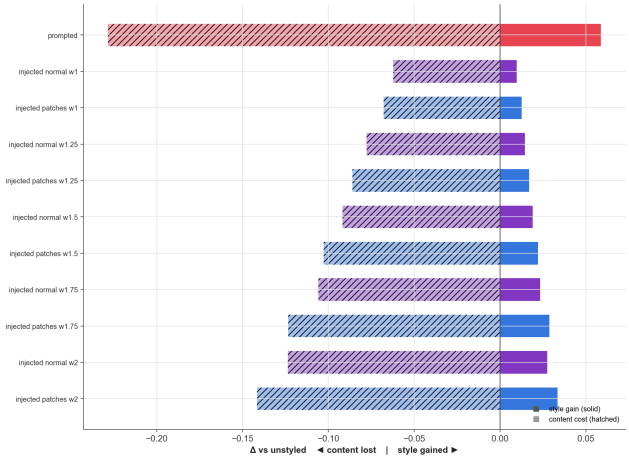


Figure 5: **Style gain vs. content cost per condition**, relative to unstyled (solid bars: `clip_style` gain; hatched bars: content cost). Prompting gains the most style but at a disproportionately large content cost; every steering condition trades a smaller content loss for a smaller style gain, and the `normal` (uniform) variant costs less content than `patches` at matched style.

guishable from prompting's observed 0.249 (Fig. 2, left). Prompting thus sits essentially on the steering frontier. The trade-off advantage is real only on the task-specific classifier (Fig. 2, right), where uniform with $w = 2$ attains higher mean `uc_style` and higher `clip_content` than prompting (0.0199 / 0.876 vs 0.0117 / 0.772). At any tested weight, steering's content cost ($\leq$ 0.124 at w=2) stays well inside prompting's 0.228, so a user can dial in partial style at a content cost lower than prompting can attain.(Fig. 5). Between the two steering variants, the uniform map (`injected_normal`) beats the patch variant on both the classifier (0.0199 vs 0.0111) and content (0.876 vs 0.858), while the patch variant edges it only on the confounded `clip_style` metric (0.224 vs 0.218; Fig. 3, 5).

6. Conclusion

We asked whether steering interpretable SAE features can apply style at controllable strength while preserving content that prompting destroys. Our results give a qualified yes, under various conditionings. We trained a top-k SAE on SD1.4's up.1.2 activations (FVU $\approx$ 0.17) and localized latents that matched styles by anchor contrast. We brought in novelty by re-injecting the decoder atoms on the conditional CFG branch to produce a smooth style–content frontier better than prompting. Overall, we saw that prompting resulted in a discrete seed-entangled switch, hard to trace. The uniform steering variant was the highest-performing method. Compared to the patch variant, uniform outperformed on both the UnlearnCanvas classifier (0.0199 vs 0.0111) and content preservation (0.876 vs 0.858), and it held clip content at 0.876 against prompting's 0.772 while reaching a styling regime. We attribute the uniform variant's edge to its broader spatial support. Summing the full style set over every row inject coherent global vector directions. Per-row Top-16 intersection in the patch variant fragments the signal and discards style latents at rows where they fall below the local cutoff.

The primary advantage over prompting sits in task-specific classification. On CLIP style, prompting sits essentially on steering frontiers. This result is confounded by CLIP rewarding alignment to the same style tokens that generated the prompted images. The two style metrics are uncorrelated (r = 0.007(, so we report both rather than privilege one. The clearest finding is that style lives in a compact, near mono-semantic feature set for some styles (Watercolor steers cleanly while content holds) and not for others (Pencil Drawing never renders).

This investigation allows for several research directions. Multi-block steering is the most immediate future work. Since up.1.2 cannot render line-dominated styles like Pencil Drawing, training SAEs across the up-sampling path would test whether those cues live elsewhere. More compute would scale the expansion factor and sweep k to split into cleaner style atoms. More data and a classifier calibrated to steered base SD1.4 would improve the uc style intervals that currently leave the advantage insignificant. Extensions to SDXL and rectified-flow backbones would test whether sparse style latents are specific to SD1.4 or gener-

alizable to diffusion frameworks as a whole.

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Contributions

All authors contributed in equal capacity to the conceptualization and execution of the project. A.N. functionalized stable diffusion within the codebase and performed preliminary experiments. A.M. worked extensively on developing and training the Sparse Autoencoders. S.D. conceptualized the experiments and wrote scripts to visualize data. All authors contributed to development of this manuscript.

AI was used task-specifically in this work, specifically for formatting the LaTeX manuscript, writing matplotlib code for figure generation, and extensively for debugging. The conceptualization, literature, and model development, however, was carried out entirely by the authors.